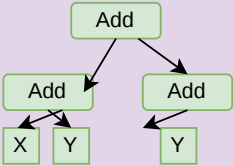


Lambda Function Support in SQL(DuckDB)

Example:SQL with lambda

```
Select mygrad_rev(  
  lambda x , y: x*y + sin(y),  
  2.0, -- x  
  0.5 -- y  
) AS (df_dx, df_dy);
```

Lambda AST



Bound Lambda

Parameters :[x, y]
Body : Add(..)
Types : [Double,
Double]
Captured Columns: []
Function Id : 42

Lambda Binding Process (During Bind Phase)

1. Parse Lambda Expression
2. Build AST
3. Resolve References
4. Type Check
5. Bind to Function
6. Compile to Executable (Tape/Ops)

Used During Execution (Per Vector /Chunk)

Input Columns (Vector) → Evaluate Lambda(Forward) → Compute Gradients (Forward/Reverse) → Output Gradients (Vector)